ADVANCING THE FRONTIERS OF EARTH SYSTEM PREDICTION

AFESP 2026

CONFERENCE PROGRAMME

Monday 2 & Tuesday 3 November 2026

Henley Business School, Whiteknights Campus
University of Reading, RG6 6UR

Conference at a glance

Two days of research presentations, keynote talks, poster sessions and collaborative discussion across the AFESP community.

DAY 1 • MONDAY 2 NOVEMBER

09.30 arrival | Welcome and programme update | Research presentations | Keynote by Dr Chris Roberts | Poster session | Breakout sessions

DAY 2 • TUESDAY 3 NOVEMBER

09.00 arrival | Research and DTP presentations | Poster session | AFESP Fellows keynote session | Open discussion

Practical information

Main auditorium	G11 Lecture Theatre
Breakout rooms	G10 and 108
Registration, refreshments and posters	Foyer
Campus address	Henley Business School, Whiteknights Campus, Reading RG6 6UR

Programme notes

- Times use the 24-hour clock.
- Please be sure register on arrival and collect your name badge
- Refreshments and lunch are served in the foyer unless otherwise stated.

DAY 1

Monday 2 November

Welcome, programme update, research presentations and keynote

TIME	SESSION
09.30-10.00	**Arrival and Registration** • Foyer Tea and coffee
10.00-10.30	**Welcome** • G11 Lecture Theatre Overview of AFESP and progress to date | **Pier Luigi Vidale**
10.30-11.30	**Research Project Presentations** • G11 Lecture Theatre 1. Towards a high-fidelity Integrated Forecasting System via improved modelling of the soil-plant system | **Rajsekhar Kandala** & Anne Verhoef 2. The ECMWF Land Surface Hydrology Foundation model (ECFoundLand) | **Hamidreza Mosaffa** & Hannah Cloke 3. UrbaNE: Urban Numerical Weather Prediction Expedited | **Cameron Southgate-Ash** & Sue Grimmond
11.30-11.45	**Refreshment Break** • Foyer Tea and coffee
11.45-12.45	**Keynote Session** • G11 Lecture Theatre **Dr Chris Roberts, Senior Scientist, European Centre for Medium-Range Weather Forecasts (ECMWF)** Progress, challenges, and "state of the art" in subseasonal prediction at ECMWF
12.45-14.00	**Lunch** • Foyer

Keynote focus

DR CHRIS ROBERTS

This talk will cover the recent evolution of sub-seasonal prediction at ECMWF, including the development and performance of fully coupled Earth System Model ensemble prediction systems and the exploration of new data-driven approaches that leverage the rapid evolution of machine learning methods for numerical weather prediction.

DAY 1 CONTINUED

Monday 2 November

Posters, breakout discussions and summaries

TIME	SESSION
14.00-15.00	**DTP Poster Session** • Foyer Posters presented by all DTP students
15.00-15.15	**Refreshment Break** • Foyer Tea and coffee
15.15-16.30	**Breakout Sessions** • Rooms G10 & 108 Two concurrent discussion sessions, with groups rotating between topics. Chairs and co-chairs will capture key ideas, views and feedback. Session 1: AFESP further opportunities and future focus for research and investment. Session 2: AFESP community HPC usage and needs.
16.30-17.00	**Breakout Session Summaries** • G11 Lecture Theatre

Breakout Session Rotation

SESSION	CHAIR (CO-CHAIRS)	ROOM	GROUP TIMES
1. AFESP further opportunities and future focus for research/investment In the first five years of AFESP, we have invested in three streams: i) PhDs, ii) PI-led Research Projects and iii) Independent Fellowships. Questions: • Which of these streams would you like to see prioritized in the second funding cycle, and why? • Are the three AFESP science themes still relevant, or do they need re-visiting/re-calibration in the next 5-year cycle? • How can we leverage the initial UoR investment in AFESP, and the new capabilities in the partnership, to seek additional funding? • Are there some partners missing from AFESP that should be brought in, and how will they increase our capability?	Pier Luigi Vidale (Mark Muetzelfeldt & Julia Kukulies)	G10	Group A 15.15-15.50 Group B 15.55-16.30
2. HPC usage and needs AFESP has provided multiple solutions for access to the latest prediction products, tools etc. Questions: • Which of these access routes have worked / not worked, and why? • What new data sets / products / tools would you like AFESP to exploit in the future (example: Destination Earth, EarthCARE) • Which new capabilities do AFESP investigators need to develop/master in the next five-year cycle? • How do any new science directions from Session 1 translate into new capability and infrastructure requirements? (example: are there new data sets that we should host/maintain directly?) • Is there a requirement to make AFESP data and tools (e.g. code) reproducible, and how can we govern and maintain that?	Sharar Ahmadi (Charlie Egan & Rishabh Bhatt)	108	Group B 15.15-15.50 Group A 15.55-16.30

DAY 2

Tuesday 3 November

Research project and DTP presentations, lunch and posters

TIME	SESSION
09.00-09.30	**Arrival** • Foyer Tea and coffee
09.30-09.40	**Day 2 Welcome** • G11 Lecture Theatre Welcome, Day 1 recap and introduction to Day 2 | **Pier Luigi Vidale**
09.40-10.40	**Research Project Presentations** • G11 Lecture Theatre 4. High-resolution data assimilation with Spatially and Temporally Correlated Observation Errors (HiSCORE) | **Rishabh Bhatt** & Sarah Dance 5. Isolating and understanding the most predictable components and improving their representation in S2S prediction | **Charlie Egan** & John Methven 6. UPFLO: Improving understanding and modelling of convective updrafts and anvil clouds | **Deepak Gopalakrishnan** & Chris Holloway
10.40-11.20	**DTP Project Presentations - Cohorts 1 & 2** • G11 Lecture Theatre 1. Understanding the potential for month-ahead prediction for monsoon systems | **Indrakshi Mukherjee** 2. Discovering the mechanisms behind "forecast busts" | **Kaustubh Mittal** 3. Developing artificial intelligence approaches to enhance representations of turbulence in atmospheric models | **Sambit Kumar Panda** 4. S2S prediction of marine heatwaves and associated compound events | **Piyali Goswami**
11.20-11.35	**Refreshment Break** • Foyer Tea and coffee
11.35-12.45	**DTP Project Presentations • Cohorts 1 & 2** • G11 Lecture Theatre 5. Improving the efficiency of weather and climate prediction by developing mathematical methods to take long time steps | **Amber te Winkel** 6. How skilful are AI-based forecasts of monsoon weather? | **Mehzooz Nizar** 7. Where do we need high resolution in the ocean? | **Gabrielle Ching-Johnson** 8. Machine-learning-driven balance relationships for next-generation data assimilation systems | **Ravi Shankar Nemani** 9. Downstream impacts of embedded convection in warm conveyor belts | **Jake Keller** 10. Improving river streamflow forecasts using deep learning techniques | **Karan Ruparell**
12.45-13.45	**Lunch** • Foyer
13.45-14.45	**DTP Poster Session** • Foyer Posters presented by all DTP students

DAY 2 CONTINUED

Tuesday 3 November

AFESP Fellows keynote session and conference close

TIME	SESSION
14.45-15.45	**AFESP Fellows Keynote** • G11 Lecture Theatre **Dr Julia Kukulies** | Convective storms in km-scale models: linking microphysics, convective dynamics and extreme precipitation **Dr Mark Muetzelfeldt** | ALOFT: Aligning Organized Features in Coarse- and High-Resolution Simulations
15.45-16.00	**Refreshment Break** • Foyer Tea and coffee
16.00-16.45	**Closing Session** • G11 Lecture Theatre Open discussion forum chaired by Pier Luigi Vidale Summary of key outcomes and actions, next steps and conference close | **Pier Luigi Vidale**

AFESP Fellows keynote abstracts

DR JULIA KUKULIES • CONVECTIVE STORMS IN KM-SCALE MODELS

Research on cloud and precipitation processes in mesoscale convective systems, including links between organised convection, microphysics and precipitation efficiency. The talk connects this process-based work to an AFESP project evaluating the added value of kilometre-scale forecasts for convective-storm precipitation using satellite observations, traditional systems and AI-based forecasts.

DR MARK MUETZELFELDT • ALOFT

ALOFT examines whether high-resolution simulations can improve convective organisation in coarser models. Spectral nudging will align lower-resolution simulations with a kilometre-scale “source of truth”, supporting fair comparisons, identification of coarse-model biases, convection parametrisation development and machine-learning applications.

Further keynote detail

Julia Kukulies will discuss comparisons of precipitation efficiency in sub-kilometre and kilometre-scale simulations, links to cloud-ice processes and observations, and possible changes under climate change. The AFESP project will assess how global kilometre-scale forecast experiments represent updraft intensity, precipitation efficiency, storm duration and impactful extremes, and when explicitly resolving convection improves prediction.

Mark Muetzelfeldt will outline how nudging increments can reveal biases in coarse-model representations of mesoscale convective systems. Combined with automated parameter estimation, these insights will inform improvements to the CoMorph convection parametrisation in the Met Office Unified Model. The resulting increments also offer a dataset for machine learning and may support better forecasts of extreme precipitation, the Indian Summer Monsoon and the Madden-Julian Oscillation.

VENUE GUIDE

Henley Business School Floorplan

Whiteknights Campus • Reading • RG6 6UR

Ground-floor venue plan • Main sessions: G11 • Breakouts: G13 and 108 • Refreshments and posters: foyer



Room key

ROOM	USE	WAYFINDING
G11	Main conference sessions	Large lecture theatre, entrance off the foyer (past cafe)
G10	Day 1 breakout session	Ground floor, right of the foyer
108	Day 1 breakout session	First floor, far end of right corridor
Foyer	Registration, catering and posters	Central area outside G11